Software Requirements Specification

for

Computer Aided Paperless Examination System

Version 1.0

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Revision History

Name	Date	Reason For Changes	Version

1. Introduction

1.1 Purpose

This Web Application provides facility to online examination worldwide. It saves time as it allows number of students/candidates to give the exam at same time and display the results at the end of the test, so there is no need to wait for the result. It is evaluated and generated by the server. Administrator has privileges to create, modify and delete the test papers and its particular questions. User can register, login and give the test with his specific id, and can see the results as well. User registered as Recruiters can request the ADMIN to post vacancies and exam papers as per their wish. It eliminates the use of items like pen, paper, etc. CAPES is a successful, state of the art, rational, online evaluation system. It is extremely effective automated, multi-user software to help the organizations meet their varied examination needs with a competitive edge. Capes ensures scientific, rational method for conducting the examinations in addition to the requisite confidentiality and security features. Its GUI based Software with Twin Architectures, for Web and Client/Server environment provides cost, time & administrative benefits for examinations. The Knowledge & Skill Test with its comprehensive and constantly updated question bank helps assess and retain the best talents in industry. The Online Data Entry/Typing Speed Test helps clients monitor and judge accurate performances. Our mission is to make CAPES a premier technology enabled Online Evaluation and Examination software solution in the IT world.

1.2 Document Conventions

This document was prepared using the IEEE recommended practice for Software Requirements Specification.

1.3 Intended Audience and Reading Suggestions

The different types of reader that this document is intended for are developers, project managers, marketing staff, users, testers, and documentation writers. The rest of this SRS contains Overall Description, External Interface Requirements, System Features, Non Functional Requirements and Other Requirements. Suggested sequence for reading the document - begin with the overview sections and proceed through the sections that are most pertinent to each reader type.

1.4 Product Scope

Scope of this project is very broad compared to manually taken exams. They are:-

- *This app can be used in educational institutions as well as in corporate world.*
- *Can be used anywhere with basic System-settings.*
- Speed and operational efficiency.

- Simplicity (just know how to use a mouse that's all!).
- Compatibility (can run on different operating systems).
- Data security (encrypted database resides on secure server).
- User friendliness (GUI based) customized question bank.
- Instant result processing
- .Better management and fully computerized environment.
- Continuing Governments E-Governance CAPES can be widely used for Selection Procedures and Education & Training.

1.5 References

This web application has been prepared on the basis of discussion with Team members and also taken information from following website – www.google.com, www.tcyonline.com and IEEE Recommended Practice for Software Requirements Specification- IEEE STD 830-1993.

2. Overall Description

2.1 Product Perspective

CAPES, the Computer Aided Paperless Examination System, is a Software application intended to ease the process of recruitment and application of Job-Vacancies in an efficient, User-Friendly, and future-ready manner. Based on Optical memory technology, especially CDROM, voluminous optically - designed question banks of multiple choice questions and answers can be stored. With the availability of a versatile computer - communication network, examination can be conducted in a space and time independent manner for a large number of examinees while obviating the need for manpower intensive supervision and security during the conduct of the examination. With stratified random generation of the set of questions for each examinee, the methods of matching the difficulty levels for all examinees are developed along with methods for minimization of biases, item selection, credit scoring, etc.

The objective of this app is to reach and connect candidate and examiner in remote communities and conduct exams in a virtual environment online. This app will only allow the registered users to enter the test module. The various stages in the app are as follows:-

Login
System Overview
Vacancies
Test
Result

Login:
This window offers the user two choices for logging into the system according to the preset privileges - Candidate login and Administrator login. The candidate login will take the user to the user profile.
The Administrator Login will take the user to the administrator profile.

System Overview:

This window can only be accessed by the administrator. It allows the administrator to add and edit exams.

Vacancies:
The Vacancies Window allow users to view the list of vacancies mapped to the Job and their respective posters (Recruiters). The applicants can apply to their desired Job after viewing the Description-along and head to take the exam for further procedure.

Test:
This window contains all the exams candidate can give. All these exams are organized according to the categories they fall in. The test window will be different for students and corporate employees.

Result:

This window displays the result of exams the candidate has just appeared. This data will be saved and displayed in the user profile. If the score is in top-cut limit, the result will also be displayed in the rankings area of result window.

2.2 Product Functions

The functions are divided according to the user types such as:-
Administrator:-
The function of the administrator is to add/edit exams in the test module.
Candidate:-

The function of the student is to update his/her profile and give various exams.
Recruiter:-
The function of the recruiter is to add/update job profiles along with the required description for candidates to apply.

The major features of CAPES database system as shown in below ER-Diagram.

2.3 User Classes and Characteristics

The various users of this app are classified into three types:-

- *Administrator*
- *Recruiter*
- *Candidate -*
    - *Student*
    - *Corporate Employees*

This software requires the user to have characteristics such as the user should be able to communicate and write in English and should have previous experience of giving an online exam. If not, the candidate should be instructed about the basics usage of the software by authorized personnel.

a. Educational level: Users should be comfortable with the English language.

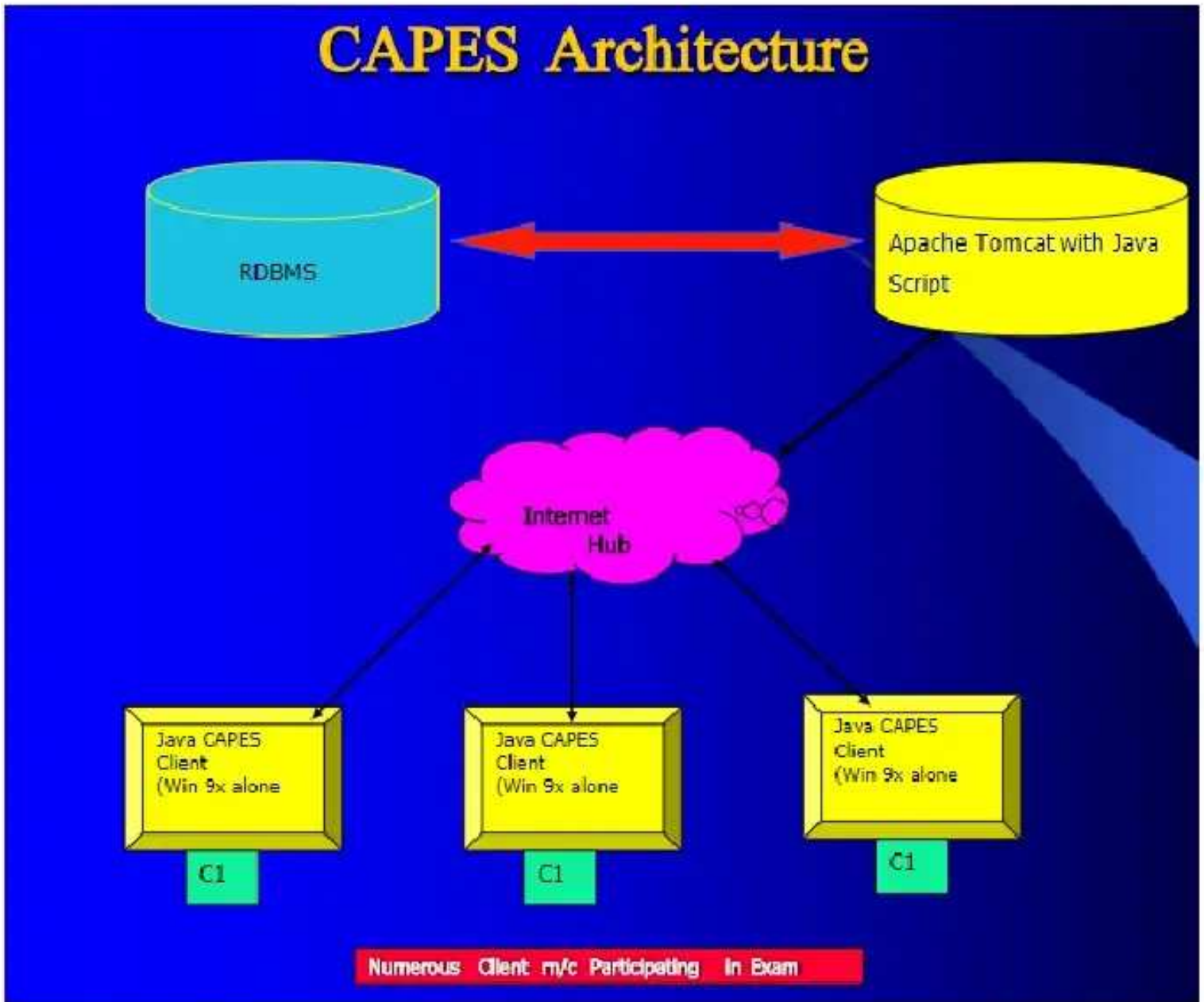
b. Experience: Users should have prior information regarding the online examinations.

c. Skills: Users should have basic knowledge and should be comfortable using general purpose applications on computers.

2.4 Operating Environment

Operating environment for the CAPES is as listed below. CAPES is designed to work on basic machine-configuration to make it an EVERYBODY's SOFTWARE. The operating Environment is intentionally made to comply with minimum and feasible technicalities.

- Distributed Database(RDBMS)
- client/server system(Apache Tomcat)
- Operating system: Windows
- database: sql+ database
- platform: vb.net/Java/javascript



2.5 Design and Implementation Constraints

- Number of Test-Takers should not be more than a certain number to account for the smooth run of the Web-server and thus avoiding Server-traffic.

- Waiting in queues for taking the test shouldn't be unfair and thus Time-stamping for the candidates should be implemented to implement fair waiting process.
- Since the CAPES uses web-server the Internet connectivity is must with other third-party software being necessarily installed on the client-machine implicitly with CAPES software.
- Extra-feature for security and fair testing should be included i.e. only the testing-software should be allowed to open and another software which can promote unfair means should be left open.

2.6 User Documentation

The product will include a user manual. The user manual will include product overview, complete configuration of the required software and hardware, technical details and contact information which will include email address.

2.7 Assumptions and Dependencies

Proper working of this software is dependent on the internet connectivity of the users' computer.

Assumptions and dependencies:

- *It is assumed that the user has basic knowledge of the system (i.e. he/she is not a first time user) as any action by the user is considered valid during an examination.*
- *It is assumed that the data entered by the user while registering is true.*
- *It is assumed that the candidate does not cheat during the exam as there are no supervisors around to monitor.*
- *The examinations are all objective. Students can give each exam just once.*

3. External Interface Requirements

3.1.1 User Interfaces

•Registration Screen:
Various fields available on this screen will be:
-Candidate_id(For Candidate)
-Name(For Recruiter : Organisation)
-Password

-Age(For Candidate)
-Qualification(For Candidate)
-Resume(For Candidate)
-Recruiter_id(For Recruiter)
-HR_Name(For Recruiter)
-Functional Area(For Recruiter)
-Location(For Recruiter)

•Login Screen:
Fields available on this screen are:
- Candidate_id(For Candidate)
- Recruiter_id(For Recruiter)
-Password

•Entering Questions:
Various Fields are:
-Questions
-Options (4)
-Correct Answer

•Exam Details Screen:

Various Fields are:
-Exam Name
-No. Of Questions
-Time Limit
-Application_id
-Candidate_id

•Result Displaying Screen:
Various Fields are:
-Candidate_id
-Candidate_Name
-Job_id
-Application_id
-Marks
-Rank
-Selection_Status

•Adding Job::
Various Fields are:
-Job_id
-Job_Description
-Experience
-Salary
-Qualification

•Applying for a Job:
Various Fields are:
-Candidate_id

-Application_id
-Job_id
-Application_Date
-Application_Status

•Post a Vacancy:
Various Fields are:
-Job_id
-Number_of_Vacancies
-Application_Start_Date
-Application_End_Date
-Exam_Date

3.1.2 Hardware Interfaces

*Support for printer for printing results then and there.

**Screen resolution of at least 800X600 is required for proper and complete viewing of screens. Higher resolution will be accepted.*

3.1.3 Software Interfaces

-Any windows based operating system.
-MySql Server Database
-Php and javascript

Following are the software used for the CAPES.

Software used	Description
Operating system	We have chosen Windows operating system for its best support and user-friendliness.
Database	To save the Merit List, Recruiter Records' we have chosen SQL+ database.
VB.Net	To implement the project we have chosen Vb.Net language for its more interactive support.

3.1.4 Communications Interfaces

None

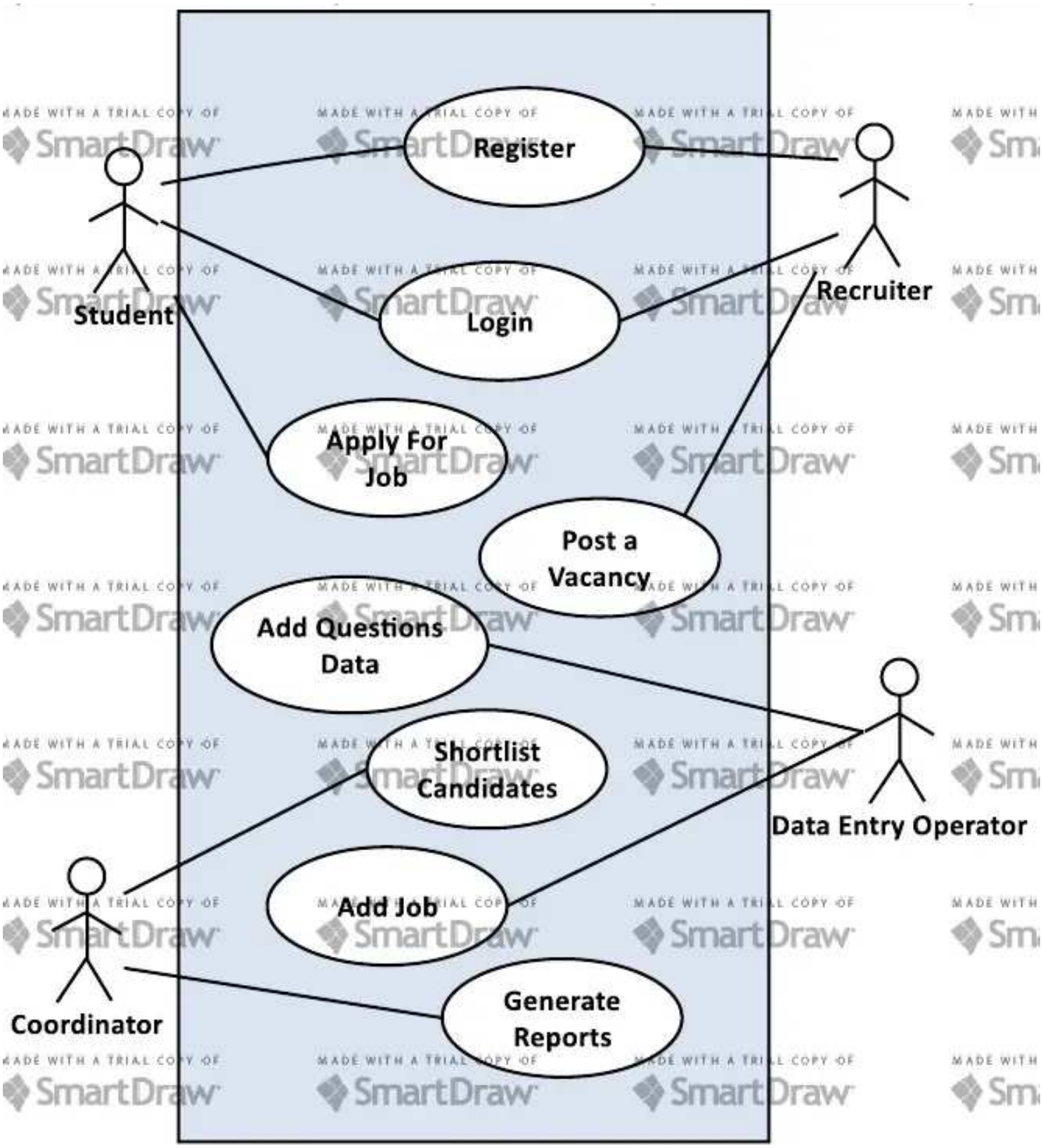
3.2 Functional Requirements

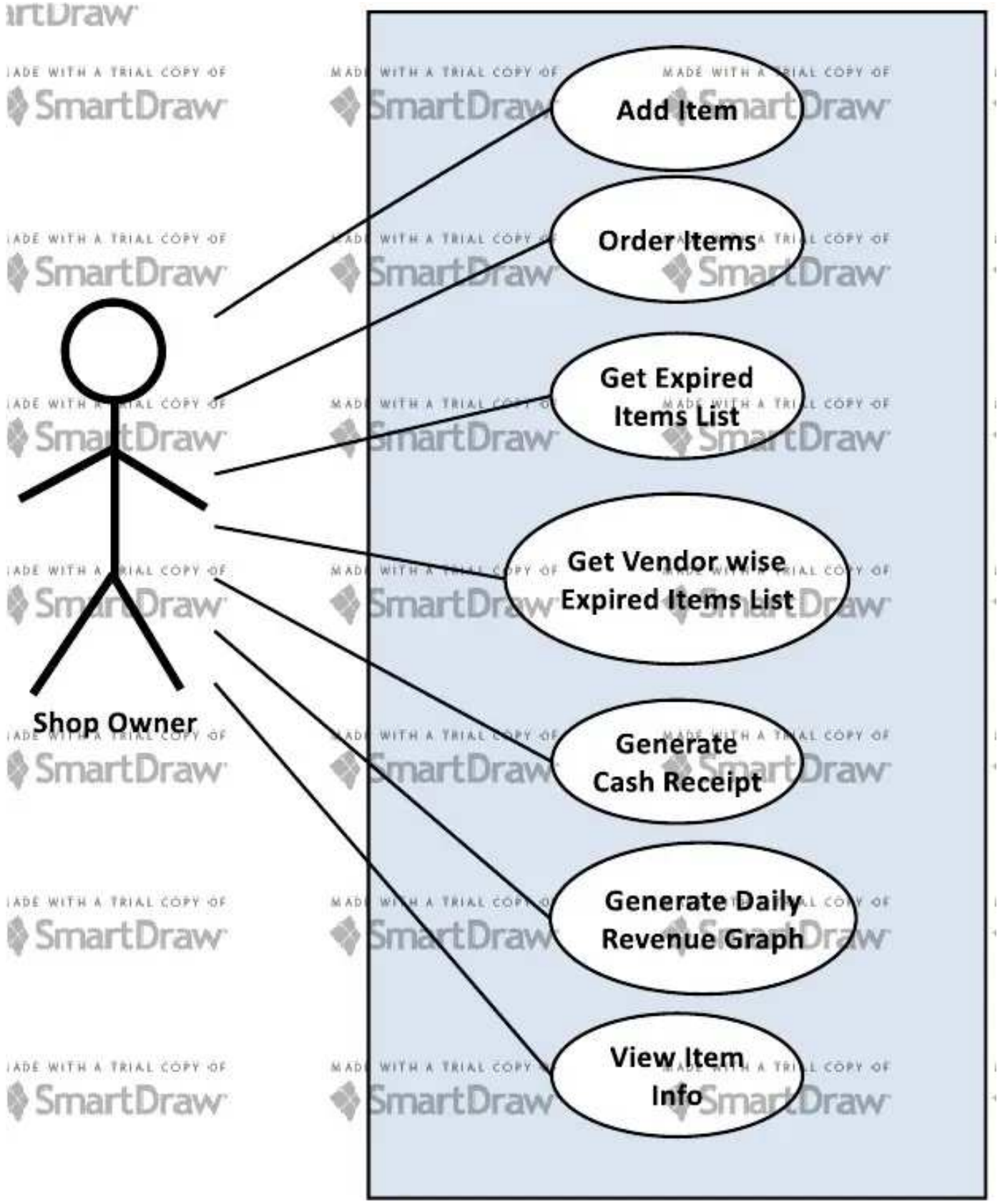
This section gives a functional requirement that applicable to the On-Line Examination system. There are three sub modules in this phase.

- *Candidate module.*
- *Examiner module.*
- *Administrator module.*
- *Result rankings module*
- *Discussion Forum module*

3.3 Behavior Requirement

3.3.1 Use Case Diagram





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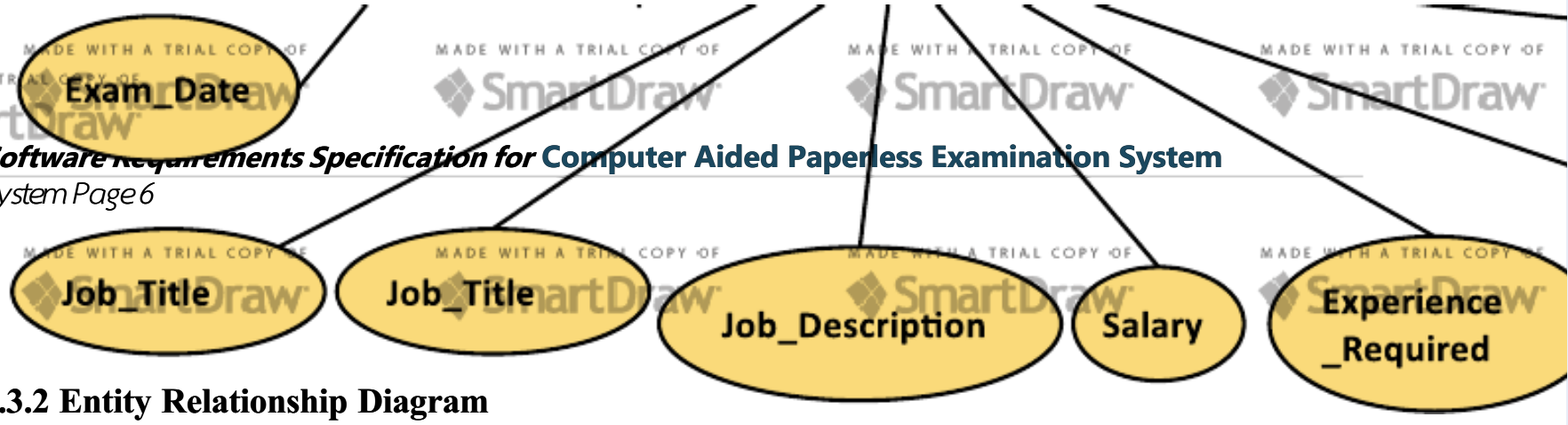
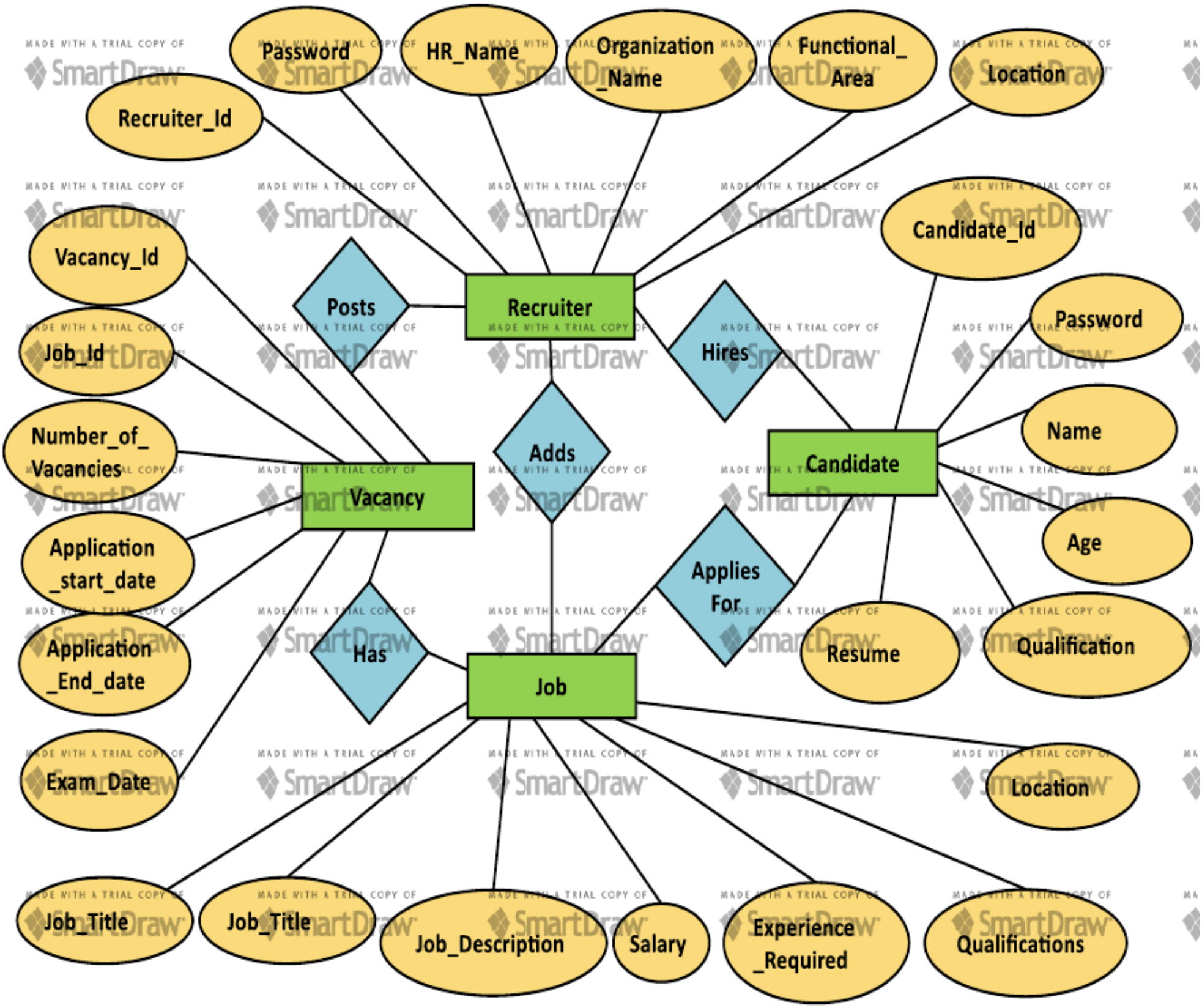
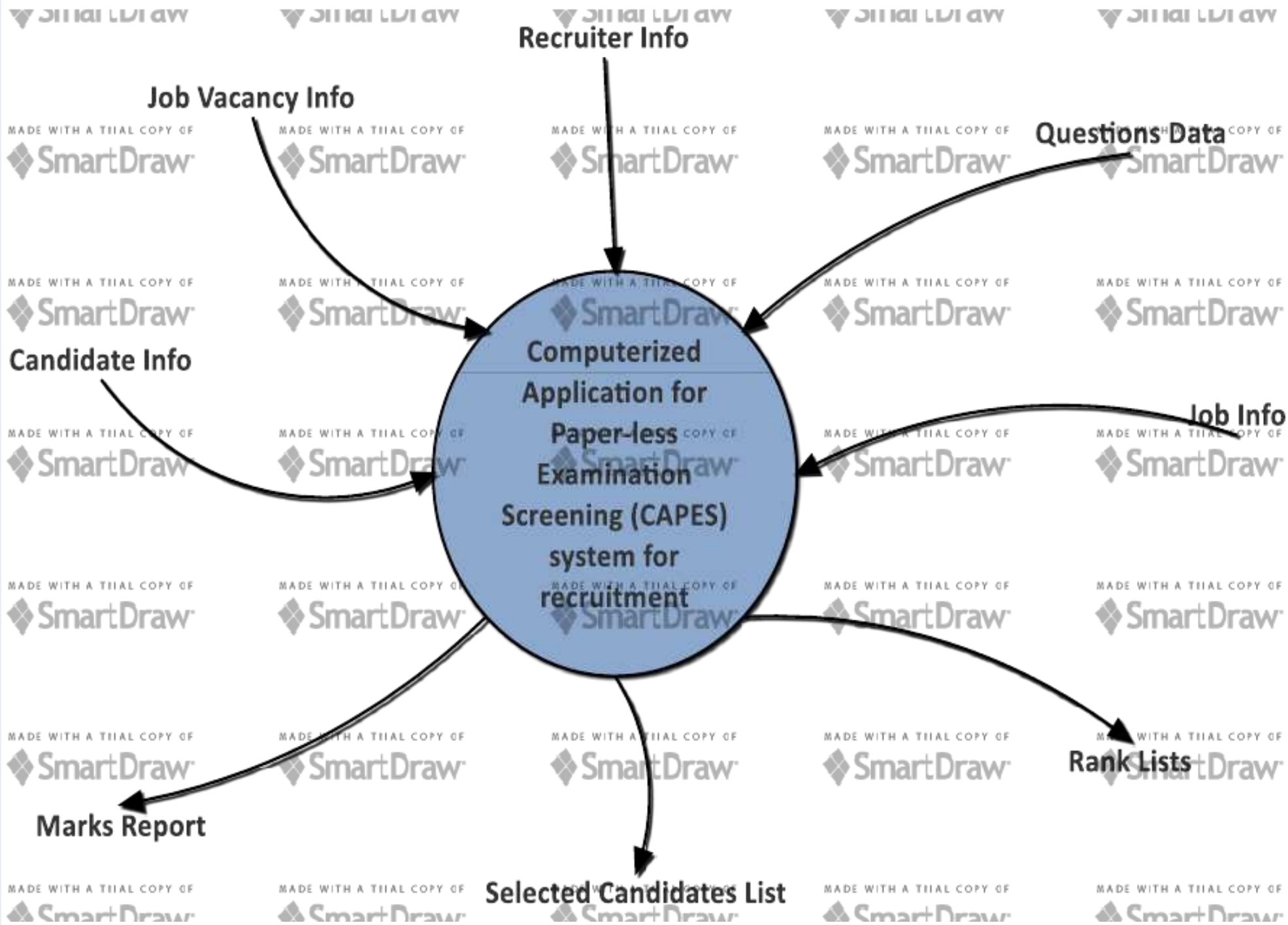


3.3.2 Entity Relationship Diagram

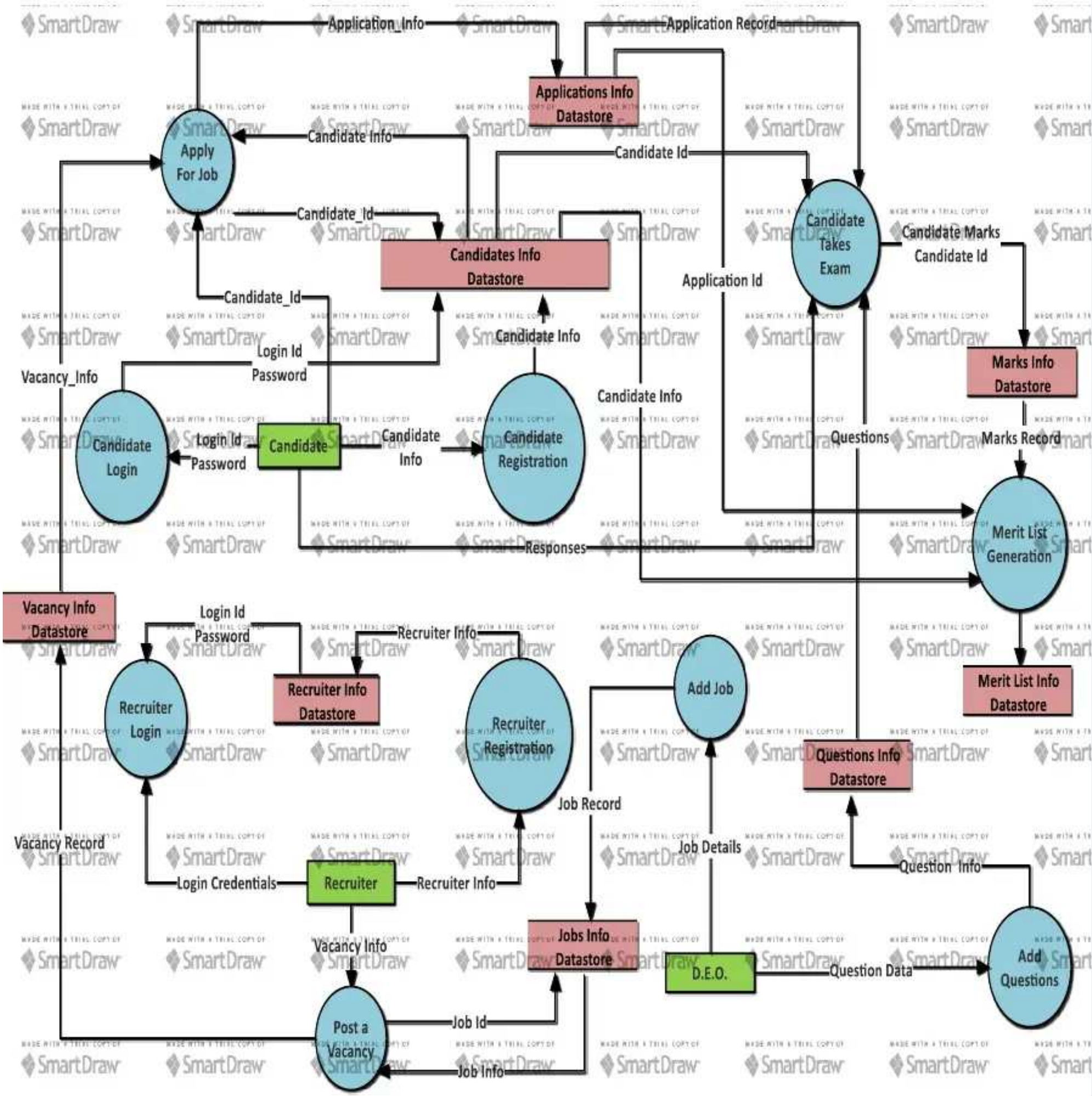


3.3.3 Data Flow Diagram

LEVEL-0 (Context Level)

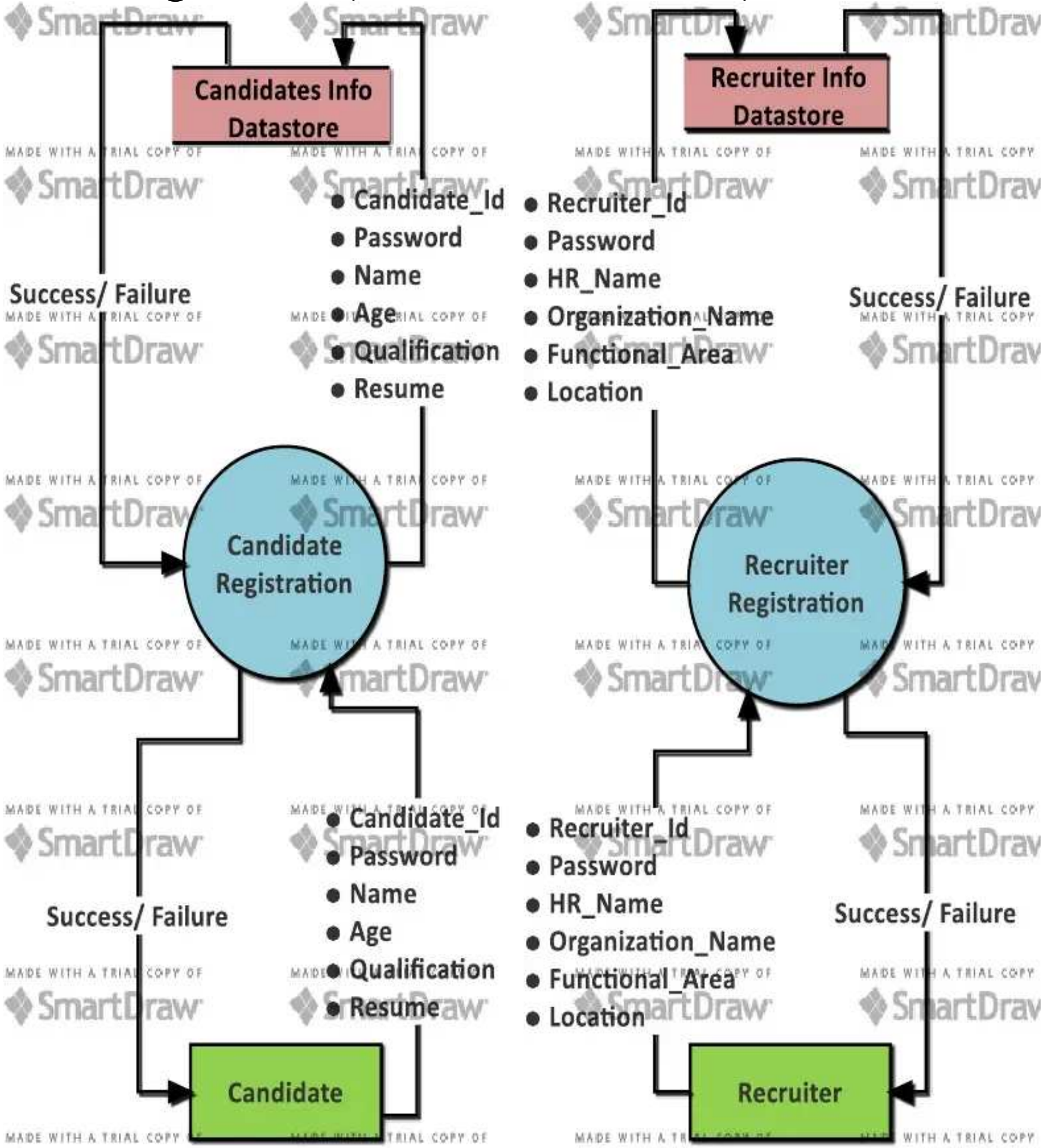


Level-1

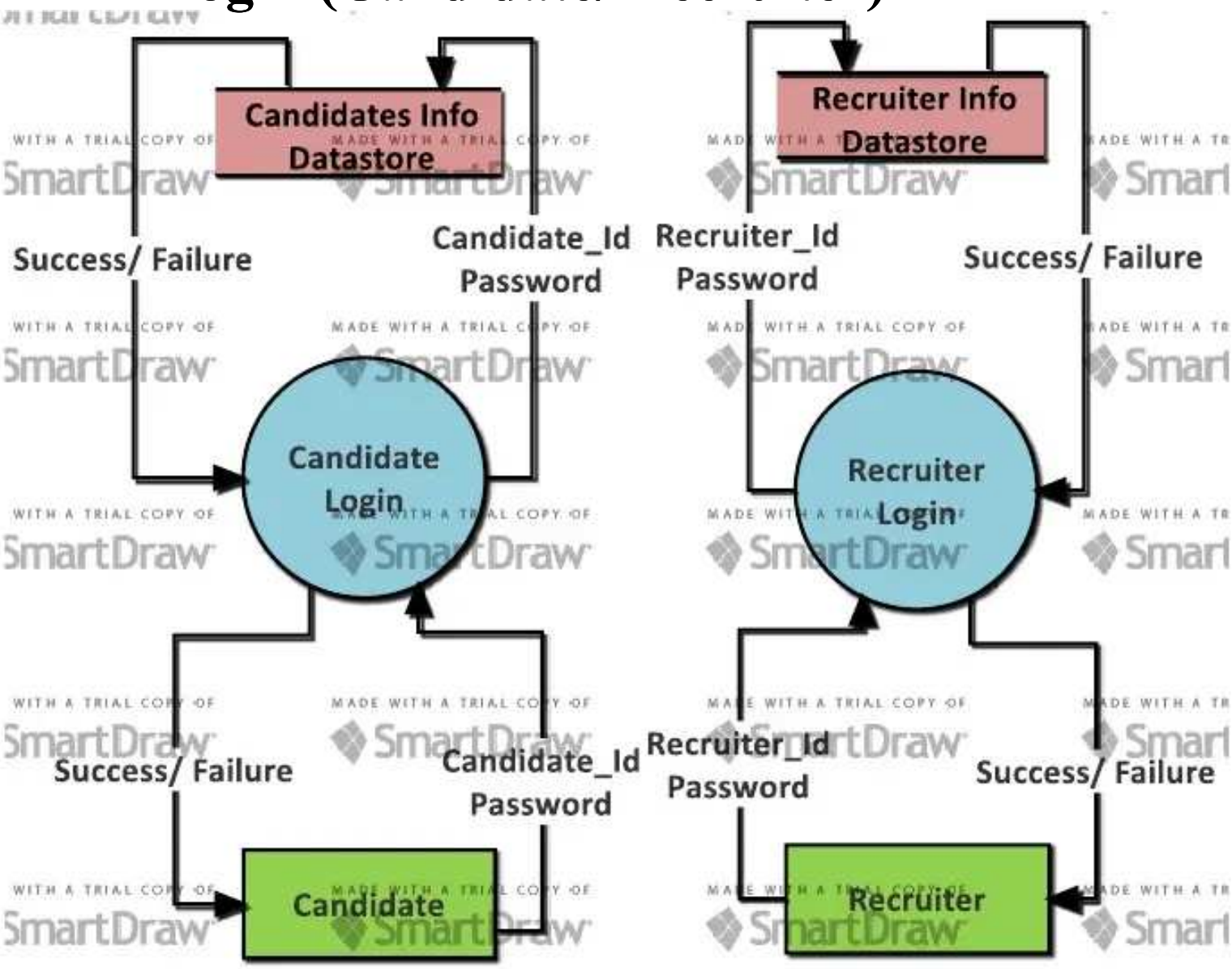

Application_Info
Application Record
Applications Info Datastore
Apply For Job
Candidate Info
Candidate Id
Candidate_Id
Candidates Info Datastore
Candidate Takes Exam
Candidate Marks
Candidate Id
Application Id
Candidate_Id
Candidate Info
Login Id
Password
Vacancy_Info
Marks Info Datastore
Candidate Info
Candidate Login
Login Id
Password
Candidate
Candidate Info
Candidate Registration
Questions
Marks Record
Responses
Merit List Generation
Vacancy Info Datastore
Login Id
Password
Recruiter Info
Recruiter Login
Recruiter Info Datastore
Recruiter Registration
Add Job
Merit List Info Datastore
Questions Info Datastore
Job Record
Vacancy Record
Job Details
Question Info
Login Credentials
Recruiter
Recruiter Info
Jobs Info Datastore
Vacancy Info
D.E.O.
Question Data
Add Questions
Post a Vacancy
Job Id
Job Info

Level-2

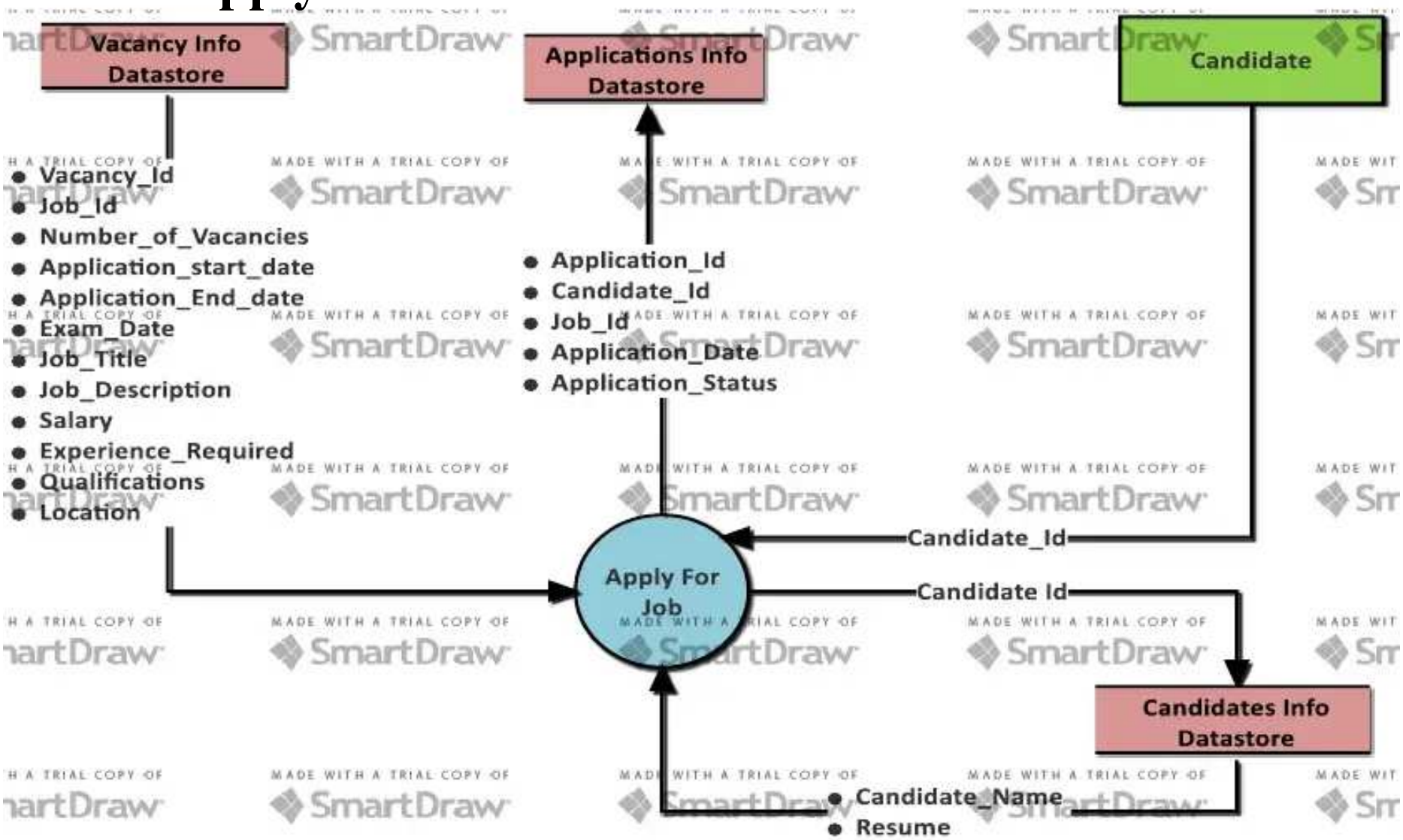
- **Registration(Candidate/ Recruiter)**



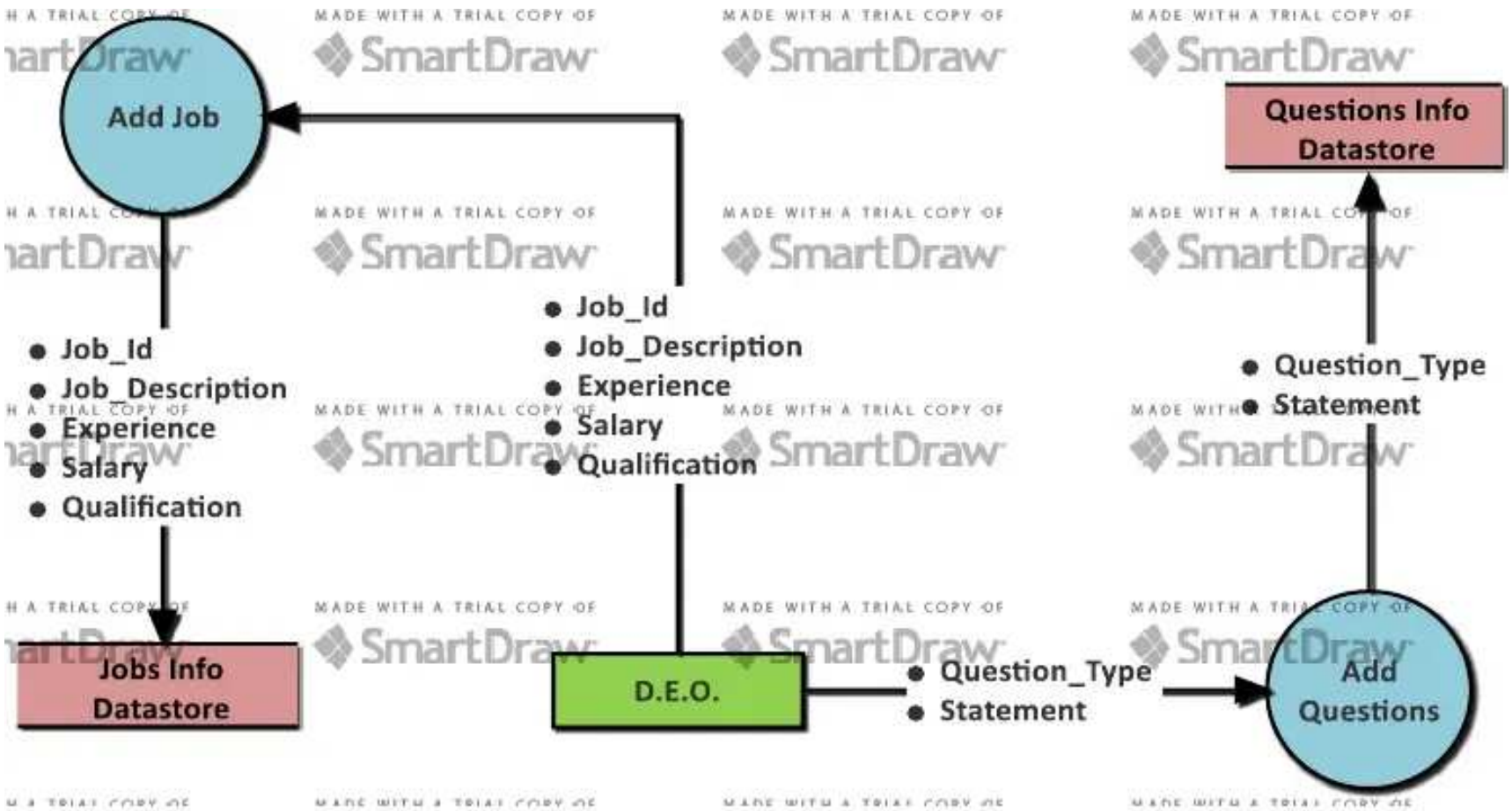
- **Login (Candidate/ Recruiter)**



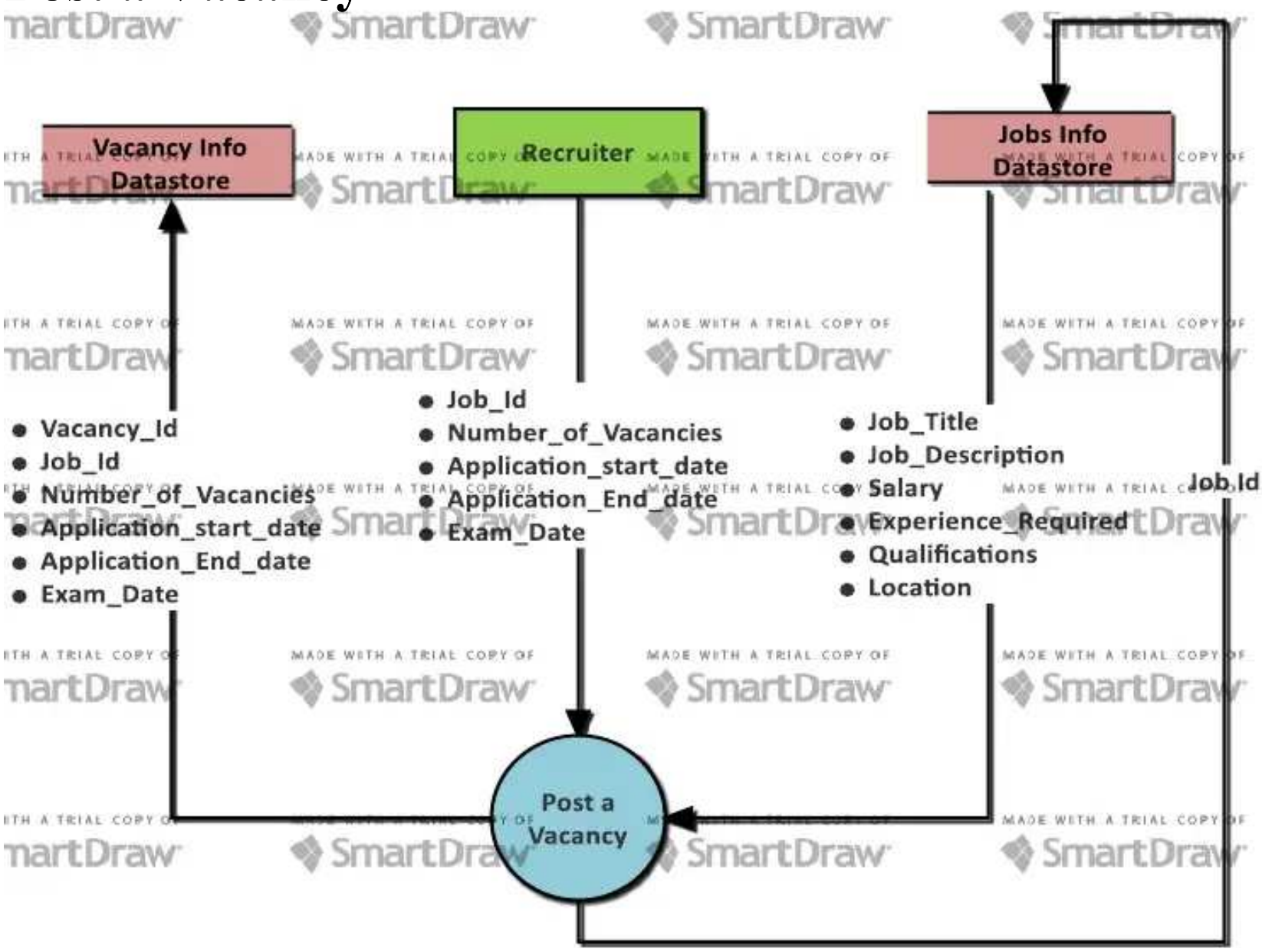
- **Apply for a Job**



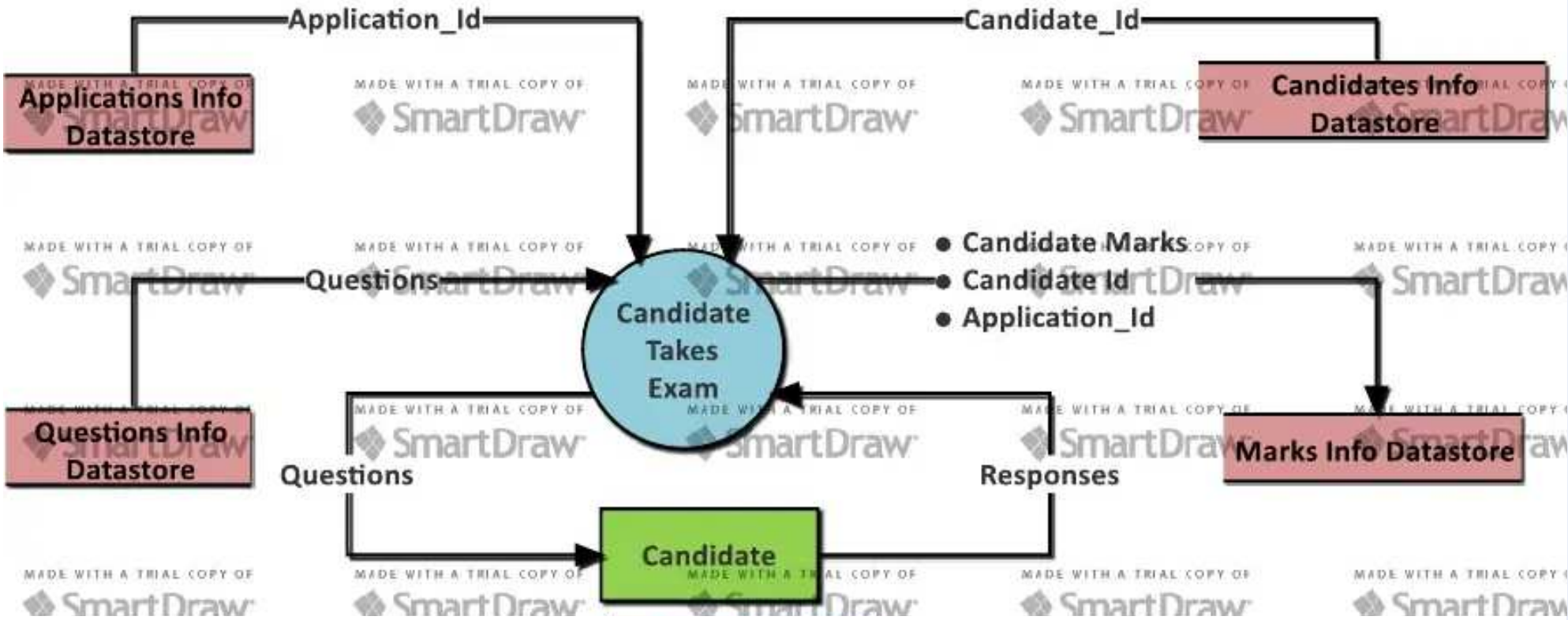
- **Add Job/Questions(Admin)**



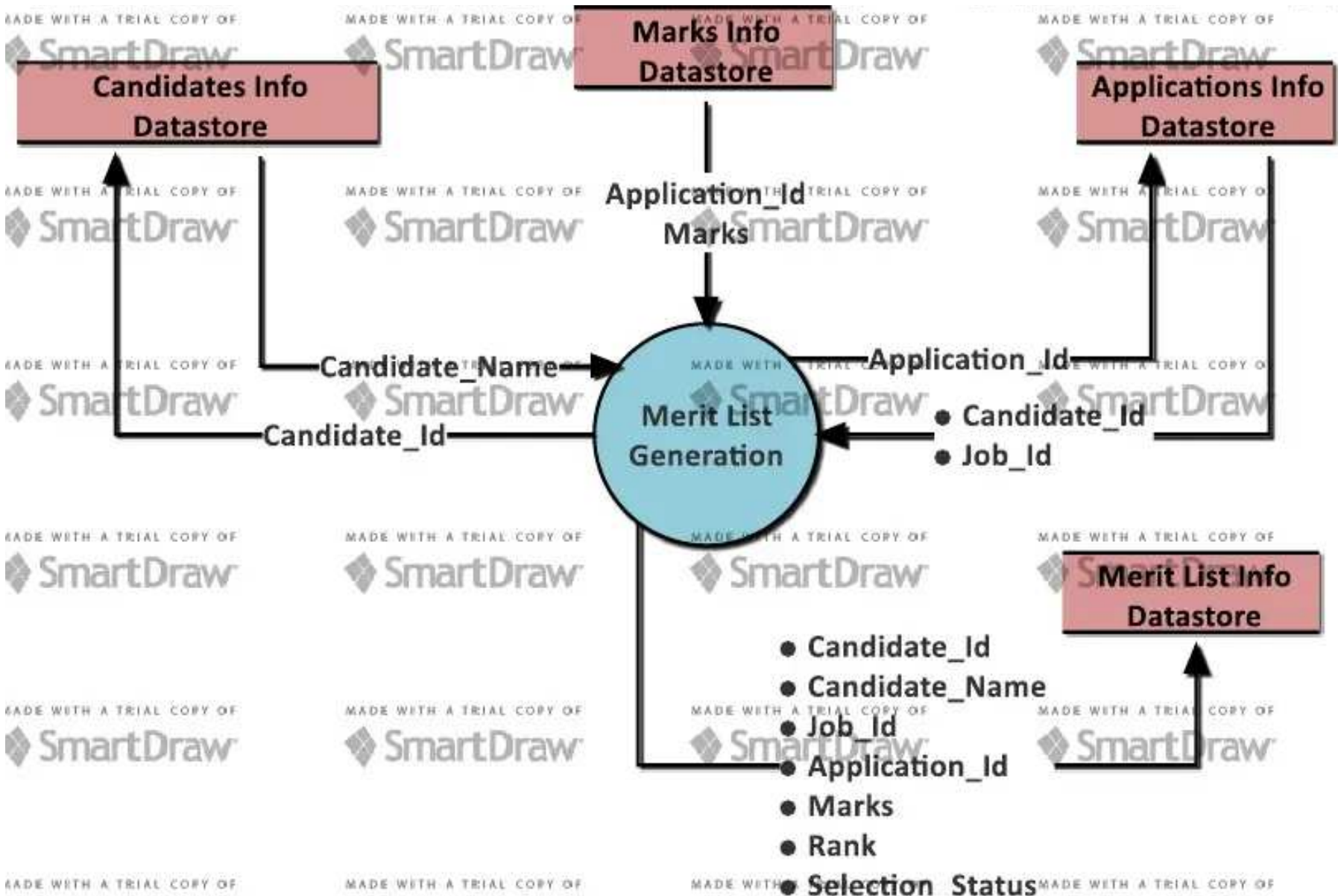
- **Post a Vacancy**



- **Candidate Takes Exam**



- **Merit-List Generation**



3.3.4Database Name: ONLINE EXAMINATION SYSTEM

Admin

Admin_ID	Name	Phone	Email	Username	Pass

User

- **Candidates Info DataStorem**

Candidate_ID	Password	Name	Age	Qualification	Res

- **Recruiter Info DataStoren**

Recruiter_ID	Password	HR_Name	Organization_Name	Functional_Area	Loc

Jobs Info DataStore

Job_ID	Job_Description	Experience	Salary	Qualification

Vacancy Info DataStore

Vacancy_Id	Job_ID	Number_of_Vcancies	Application_start_date	Application_End_date	Ex

Application Info DataStore

Application_Id	Candidate_Id	Job_Id

Question Info DataStore

Question_Type	Statement

Marks Info DataStore

Application_Id	Candidate_Id

Merit List Info DataStore

Application_Id	Job_ID	Candidate_Id	Candidate_Name	Marks	Rank	Selection_Status

4. System Features

Some Performance requirements identified is listed below:

- *The database shall be able to accommodate a minimum of 1,000 records of Users.*
- *The software shall support use of multiple users at a time.*
- *There are no other specific performance requirements that will affect development.*

4.1 Safety and Security Requirements

4.1.2 Safety Requirements

The database may get crashed at any certain time due to virus or operating system failure. Therefore, it is required to take the database backup.

4.1.3 Security Requirements

Some of the factors that are identified to protect the software from accidental or malicious access, use, modification below. Keep specific log or history data sets

- Assign certain functions to different modules

4.2 Software Quality Attributes

The Quality of the System is maintained in such a way so that it can be very user friendly to all the users.
The software quality attributes are assumed as under:

- Accurate and hence reliable.
- Secured.
- Fast speed.
- Compatibility.

5. Other Requirements

Display result on user's email-id.

Appendix A: Glossary

Appendix B: Analysis Models

Appendix C: To Be Determined List